Toward Adaptive HyRE for Legal RAG: Reasoning-Conditioned Retrieval Across Legal Tasks

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Abstract

Legal retrieval-augmented generation is usually summarized as one accuracy number, but legal QA failures are not one phenomenon. We study HyRE (Hypothetical Reasoning Embeddings), a snap-first variant of HyDE in which a model first reasons toward an answer and then generates the hypothetical retrieval text from that reasoning. We evaluate HyRE across BarExamQA (bar questions), HousingQA (state statutes), LegalBench-SCALR (holding selection), and CaseHOLD (holding options), using one full-corpus legal result and paired $N = 200$ diagnostics to probe RAG bottlenecks. On full-corpus BarExamQA, `rag_snap_hyde` improves Gemma 4 26B from 78.08% to 81.17% (+3.09pp, McNemar $p = 0.0130$) and Gemma 4 E4B from 58.49% to 62.18% (+3.68pp, $p = 0.0129$). The remaining evidence is heterogeneous in a structured way: SCALR is candidate-depth sensitive then saturates at top-5, HousingQA improves with metadata-filtered retrieval, and CaseHOLD improves gold-option retrieval without yet converting that into a reliable answer lift. We interpret this heterogeneity as evidence for adaptive HyRE: a controller that places snap-first reasoning on the active bottleneck. The resulting agenda is generalizable in form, even where current experiments are $N = 200$: diagnose whether a legal task needs more candidates, better query generation, answer anchoring, metadata filtering, or stronger evidence-to-option conversion, then spend the reasoning budget there.

1 Introduction

Legal QA is a natural target for retrieval augmentation. Many questions depend on statutes, holdings, jurisdiction, procedural posture, and precise rule exceptions that may not be reliably stored in a model's parameters. However, legal RAG is also a setting where more retrieved text can easily hurt. A passage can be topically related but legally incomplete; a holding can mention the same doctrine while being the wrong option; a state statute can be right in subject matter but wrong in jurisdiction. These errors are not just retrieval failures or generation failures. They are interactions between retrieval, reasoning, and answer selection.

Our initial project pursued a broader agentic legal RAG system: route the question, decompose the issue, retrieve per gap, judge sufficiency, and replan. That design matches legal research practice, but it is difficult to evaluate scientifically because any gain could come from retrieval, extra tokens, a second reasoning pass, or prompt luck. We therefore moved from full-system complexity to atomic retrieval interventions.

The central method family is HyRE. A "snap" is a first-pass answer or reasoning trace. HyRE then generates a hypothetical legal passage from that snap and embeds it for retrieval. The distinction from standard HyDE is that the model's reasoning about how to solve the question is baked into the hypothetical snippet before retrieval happens. In the core sequential and two-call modes, the final answer sees the question and retrieved evidence; the snap shapes retrieval rather than being handed directly to final synthesis. Separate snap-only and one-call ablations test what happens when visible snap reasoning is itself part of the final answer path. The intuition is legal: before asking the model to read external authority, ask it to state the issue and likely rule. Retrieval is then shaped by that legal frame rather than by the raw question alone.

The current legal-only evidence varies by task, but it varies in a structured way. BarExamQA shows a real full-corpus HyRE lift, but top-1 versus top-5 retrieval is flat, so the lift should not be described as "more documents help." LegalBench-SCALR shows a strong top-1 collapse and top-5 saturation, which suggests candidate-set size matters more than pseudo-document query formulation. HousingQA shows that jurisdiction-aware retrieval can outperform generic retrieval, reaching 62.5% at k=10. CaseHOLD exposes a retrieval-to-answer gap: HyRE raises gold-option retrieval from 16.0% to 47.0%, but answer accuracy only moves from 69.5% to 72.0% with $p = 0.4421$.

The forward hypothesis is that HyRE should become adaptive: it should choose whether to spend its reasoning budget on query framing, query diversity, candidate expansion, metadata filtering, or answer-option conversion depending on what the problem appears to need. This is the generalization claim we can defend now: not that one prompt wins every benchmark, but that the limiting stage of a legal RAG pipeline can often be diagnosed and targeted directly.

2 Related Work

Retrieval-augmented generation combines parametric generation with retrieved non-parametric evidence [Lewis et al., 2020]. HyDE generates a hypothetical document and embeds it for zero-shot retrieval [Gao et al., 2023], while Query2Doc expands queries with LLM-generated pseudo-documents [Wang et al., 2023]. FLARE retrieves during generation by using predicted future content as a query [Jiang et al., 2023]. Our method family inherits directly from this line: generated text is used as a retrieval handle, not as trusted evidence.

Adaptive RAG work asks when or how retrieval should be invoked. Self-RAG learns to retrieve and critique through reflection tokens [Asai et al., 2024]. Corrective RAG grades retrieved evidence and triggers correction when retrieval is unreliable [Yan et al., 2024]. Adaptive-RAG routes questions by complexity [Jeong et al., 2024]. Speculative RAG drafts multiple answers from retrieved subsets and verifies them with a larger model [Wang et al., 2025]. These systems motivate the main design lesson we adopt: retrieval should be conditional on observed failure mode.

Legal NLP benchmarks increasingly show that legal RAG cannot be reduced to generic open-domain QA. LegalBench provides a broad suite of legal reasoning tasks, including SCALR holding selection [Guha et al., 2023]. CaseHOLD evaluates whether a model can select the holding referenced by a court-opinion context [Zheng et al., 2021]. Zheng et al. introduce BarExamQA and Housing Statute QA to evaluate realistic legal retrieval and answering workflows [Zheng et al., 2025]. LegalBench-RAG focuses on retrieving minimal relevant legal snippets [Pipitone and Alami, 2024], and Legal RAG Bench provides an end-to-end legal RAG benchmark with factorial retriever/generator decomposition [Butler and Butler, 2026]. RAGChecker similarly argues for fine-grained separation of retrieval and generation errors [Ru et al., 2024]. Our work follows this diagnostic turn: the same RAG method should be judged differently depending on whether the active failure is retrieval depth, candidate-set selection, metadata filtering, or evidence conversion.

HyQE ranks contexts with hypothetical query embeddings [Zhou et al., 2024]. HyRE follows the same broader pattern that generated representations can serve as retrieval handles, but it makes the representation explicitly reasoning-conditioned: the hypothetical snippet is generated after a snap answer, so the embedding reflects a proposed legal solution rather than a generic pseudo-document.

3 Framework Design

3.1 Method ladder

The report uses an inheritance-style method ladder. Each method adds one capability where possible, so negative results remain interpretable. Call counts refer to LLM generation calls per question; retrieval, embedding, and reranking are not counted. Thus top-k and state-filter comparisons keep the same one-call answer budget, while HyDE and HyRE spend extra calls to generate retrieval text or snap reasoning.

Method	Calls	What it isolates
`llm_only`	1	Parametric legal answer without retrieval.
`snap_only_in_final`	2	Whether a first-pass answer/reasoning trace helps without retrieval.
`golden_passage`	1	One labeled passage supplied as a diagnostic control, not deployable retrieval.
`rag_simple`	1	Standard retrieval plus answer generation.
`rag_state_filter`	1	Standard retrieval constrained by jurisdiction metadata before final answer generation.
`rag_hyde`	2	Question-conditioned hypothetical document retrieval.
`rag_snap_hyde`	3	Sequential HyRE retrieval plus final synthesis.
`rag_snap_hyde_1call`	1	Ordinary retrieval, then one answer call that includes snap-style reasoning.
`rag_snap_hyde_2call`	2	Fixed-cost snap and HyRE passage in one first call, final synthesis in a second.
`multi_hyde_diverse`	2	Multiple generated retrieval passages for query diversity.
`subagent_rag`	var.	Gap-routed report generation; currently used as a negative over-abstention control.

3.2 HyRE: snap-conditioned hypothetical reasoning

Figure 1 shows the core method. The model first writes a snap answer or legal analysis. It then writes a hypothetical passage that states the likely controlling rule, fact, or holding. That passage is embedded and used for retrieval. The final model sees retrieved passages along with the original question; the snap is mediated through the retrieval query unless the run is an explicit snap-visible ablation.

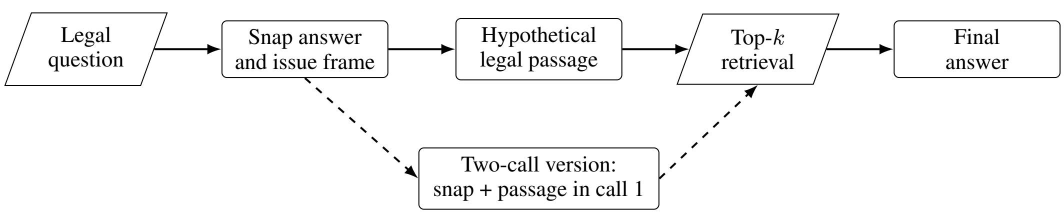

Figure 1: HyRE retrieval. The generated passage shapes retrieval; the retrieved corpus passages remain the external evidence used by final synthesis.

3.3 Bottleneck diagnostics

We use "bottleneck" to mean the pipeline stage where extra effort most changes the final answer. Three diagnostics locate that stage. First, top-k sensitivity tests whether answer accuracy depends on retrieving more than the single best passage. This is a retrieval-policy stress test, not pure context truncation, because the cross-encoder reranks a first-stage candidate pool before selecting the final top-k. Second, snap-only and HyDE-only ablations ask whether the gain comes from answer anchoring, generated-query retrieval, or their combination. Third, gold-retrieval and disagreement analyses ask whether better retrieval actually converts into better final answers.

The four legal benchmarks map naturally onto different bottleneck hypotheses: BarExamQA tests answer anchoring under strong legal priors, HousingQA tests state-specific legal retrieval, LegalBench-SCALR tests small candidate-set holding selection, and CaseHOLD tests whether a retrieved holding can be converted into the correct answer option. For the newer benchmarks, $N = 200$ diagnostic subsets let us compare mechanisms under a common evidence budget before scaling the most informative routes.

4 Experimental Setup

Benchmarks. BarExamQA contains 1,195 bar-exam multiple-choice questions with fact patterns and legal passages. HousingQA contains yes/no housing-law questions over state statutes. LegalBench-SCALR contains 571 Supreme Court holding-selection questions. CaseHOLD contains court-opinion contexts and five candidate holdings.

Models. The full-corpus BarExamQA experiments use Gemma 4 26B-A4B and Gemma 4 E4B. LegalBench-SCALR and CaseHOLD use Llama 70B, and HousingQA uses Gemma 4 26B. Paired comparisons use the same model and question subset. Except for BarExamQA, results should be read as diagnostic subsets rather than model-independent benchmark rankings.

Metrics. We report accuracy for multiple-choice and yes/no datasets. Paired deltas use exact McNemar tests with b/c discordance counts. Retrieval diagnostics report gold retrieval when the benchmark provides aligned gold identifiers.

5 Results and Analysis

5.1 BarExamQA: full-corpus HyRE lift

BarExamQA is the strongest legal positive result. On Gemma 4 26B-A4B, `rag_snap_hyde` reaches 81.17%, compared with 78.08% for `rag_simple` (+3.09pp, b/c=124/87, $p = 0.0130$). On Gemma 4 E4B, the same method improves 58.49% to 62.18% (+3.68pp, b/c=172/128, $p = 0.0129$).

Table 1: BarExamQA full-corpus Gemma 4 26B-A4B method matrix, $N = 1195$.

Method	Accuracy	Δ vs `rag_simple`
`rag_simple`	78.08%	–
`golden_passage`	78.66%	+0.58pp
`rag_hyde`	78.91%	+0.83pp
`llm_only`	79.75%	+1.67pp
`snap_only_in_final`	80.59%	+2.51pp
`rag_snap_hyde`	**81.17%**	**+3.09pp**

The mechanism is not pure retrieval depth. On a paired $N = 200$ diagnostic, `rag_simple` top-5 is 82.5% and top-1 is 83.0% ($p = 1.0$). The full-corpus lift is consistent with snap-conditioned answer framing plus retrieval, with model-size dependence: the 26B result suggests much of the gain may come from strong first-pass legal priors rather than retrieval depth alone, while retrieval/synthesis appears more useful on the smaller E4B model. A legal MC model can have strong prior knowledge of the rule; retrieval then helps only if it is framed in a way that does not overwrite the right legal analysis with a partial snippet. Figure 2 shows this cross-size pattern.

5.2 Four-benchmark legal bottleneck matrix

Table 2 is the compact result summary. It intentionally includes negative and directional results because those results identify where the next adaptive route should spend budget. The current evidence is therefore more promising as a bottleneck map than as a single leaderboard: it shows which legal bottleneck each dataset is probing.

SCALR and CaseHOLD are both holding-selection tasks, but they fail differently: SCALR becomes much stronger once the useful candidate set is present, while CaseHOLD can retrieve the gold holding more often without reliably selecting its answer label.

Figure 3 summarizes the key diagnostic split: the same top-k or HyDE intervention has different meaning depending on whether accuracy and gold retrieval move together.

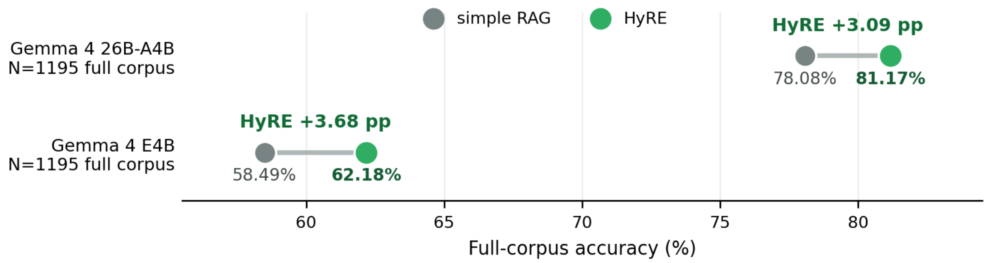

Figure 2: BarExamQA cross-size result. `rag_snap_hyde`, our HyRE implementation, improves over `rag_simple` on both Gemma 4 model sizes. The snap conditions retrieval; visible snap reasoning is tested by separate controls.

Table 2: Legal benchmark matrix.

Benchmark	Active bottleneck	Best current evidence	Caveat
BarExamQA, Gemma 4	Answer framing; retrieval depth is flat	Full $N = 1195$: HyRE is +3.09pp at 26B and +3.68pp at E4B; top-5 82.5% vs top-1 83.0% on $N = 200$	Single gold passage is a noisy control, not an oracle ceiling.
HousingQA, Gemma 4	Jurisdiction filtering	$N = 200$: top-1 50.5% to top-10 58.0% (+7.5pp, $p = 0.0722$); state filter reaches 61.5% at k=5 and 62.5% at k=10	Generic depth helps directionally; state filtering is the cleaner signal.
SCALR, Llama 70B	Top-5 candidate recall, then saturation	$N = 200$: top-1 59.5%, top-5 77.0%, top-10 77.0%; HyDE 77.5%; diverse HyDE 75.5%	Generated queries do not yet add answer lift.
CaseHOLD, Llama 70B	Evidence-to-option mapping	$N = 200$: top-1 64.5%, top-5 69.5%, top-10 68.0%; HyDE and two-call both 72.0%	Gold retrieval improves faster than answer accuracy.

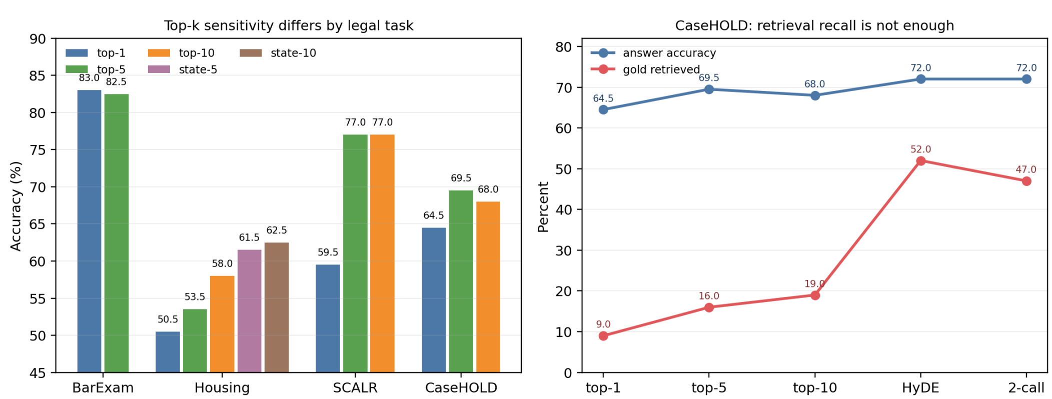

Figure 3: Legal benchmarks expose different bottlenecks. BarExamQA is depth-flat, SCALR benefits sharply from top-5 candidates, HousingQA improves with state-filtered retrieval, and CaseHOLD shows that gold-option retrieval can improve faster than answer accuracy.

The most important negative result is CaseHOLD. Top-1 `rag_simple` reaches 64.5% with 9.0% gold retrieval, while top-5 reaches 69.5% with 16.0% gold retrieval (+5.0pp, $p = 0.0525$). Top-10 retrieves more gold options (19.0%) but drops to 68.0%. HyDE retrieves the gold option in 52.0% of questions and two-call HyRE in 47.0%, but both answer at 72.0%; neither is a significant lift over top-5 `rag_simple`. If exact-gold retrieval were the only binding bottleneck, we would expect a clearer answer lift. Instead, the flat answer result points to an evidence-to-option bottleneck: retrieved holdings can still be paraphrased distractors, overly generic, or hard to map back to the exact candidate label.

SCALR gives a different holding-selection story. It is sensitive to retrieval policy: top-1 is too narrow, top-5 supplies the useful candidate set, top-10 adds recall without net answer gain, and `rag_hyde` is also flat at 77.5% despite retrieving the gold holding in 66.0% of questions. A one-call snap-final variant drops to 70.5% ($p = 0.00235$ versus top-5), so simply adding a snap rationale to the final answer is not a free improvement on this task. Diverse HyDE also stays below top-5 at 75.5%. This is not a generic "legal holdings ignore retrieval" result. It says two apparently similar holding-selection tasks diverge: SCALR needs a compact candidate set, while CaseHOLD needs better evidence conversion.

HousingQA is the strongest current evidence in this project that legally salient metadata can be a first-order retrieval variable. Generic retrieval improves from 50.5% at top-1 to 58.0% at top-10. State-filtered retrieval reaches 61.5% at k=5 and 62.5% at k=10, which is +9.0pp over generic top-5 `rag_simple` (b/c=34/16, $p = 0.015$) and +4.5pp over generic top-10 ($p = 0.306$). The k=10 state-filtered setting adds only +1.0pp over state-filtered k=5 ($p = 0.815$), so the useful signal is jurisdiction filtering rather than deeper retrieval by itself.

5.3 BarExam single gold passage is a noisy control

The BarExamQA golden-passage control appears paradoxical: providing one gold passage reaches only 78.66%, below `llm_only` at 79.75% and `snap_only_in_final` at 80.59%. The reason is legal evidence granularity. A labeled passage can be topically correct but not decisive for the fact pattern; it can also highlight a distractor doctrine. Snap-first reasoning gives the model an initial legal frame before evidence reconciliation, which partly protects against over-reading one snippet. This claim is specific to the BarExamQA single-passage control; SCALR behaves differently, with high conditional accuracy when the gold holding is retrieved. Figure 4 visualizes the paired flips behind this control.

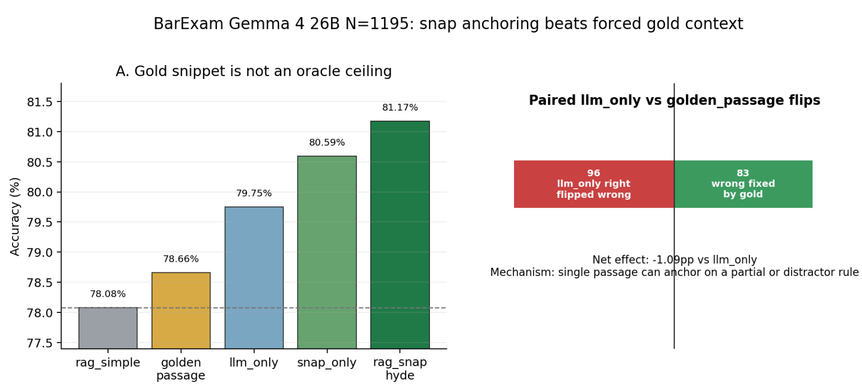

Figure 4: BarExamQA mechanism comparison. A single gold passage is a noisy control rather than an oracle ceiling; snap-first reasoning can recover some cases where forced context distracts the model.

5.4 Cost and budget behavior

The ablations also expose when extra calls and tokens are worth spending. Figure 5 plots paired $N = 200$ legal subsets with marker size proportional to total tokens per question. Same-call controls are the cleanest budget comparisons: top-k and state-filter rows keep one final answer call, while HyDE and HyRE rows spend one or two additional reasoning or pseudo-document calls. On the

$N = 200$ BarExam cost subset, two-call HyRE is the cleanest budget tradeoff: it reaches 85.5%, nearly matching sequential HyRE at 86.0%, though the paired lift over `rag_simple` is directional rather than significant. HousingQA behaves differently: generic top-10 retrieval is one call but expensive in total tokens (4.6k tokens per question), while two-call HyRE is slightly lower accuracy at a smaller 3.2k tokens per question. State-filtered retrieval is now the stronger Housing candidate because it uses jurisdiction metadata directly rather than spending generic depth; k=10 state filtering is 62.5%, only +1.0pp above k=5. The generic top-10 result also reduces a top-1 tendency to over-predict "Yes" on a dataset whose gold labels are mostly "No," so the generic depth gain should not be over-described as jurisdiction-specific retrieval. SCALR and CaseHOLD show the main caution: two-call reasoning roughly doubles calls and tokens, but SCALR does not improve and CaseHOLD improves only modestly. This is why we frame method choice as budgeted routing, not as always-on HyRE.

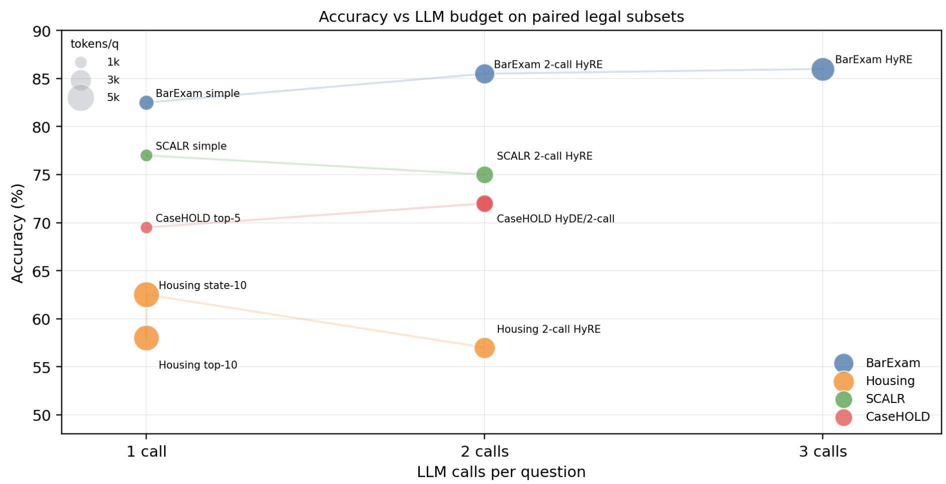

Figure 5: Calls and tokens per question matter. Bubble size is total tokens per question. Extra reasoning budget is useful on BarExamQA, ambiguous on HousingQA and CaseHOLD, and not currently justified on SCALR.

6 Discussion: From Method Search to Bottleneck Routing

The most useful interpretation of these results is not that HyRE is the final method. It is that reasoning-conditioned hypothetical generation gives RAG a place to reason about the problem before retrieval, and each dataset reveals a different way that representation step should be aimed. Sometimes the right move is a more legally framed query; sometimes it is more query diversity, a larger candidate set, a metadata filter, or an answer-option mapper after retrieval. This matters because many RAG papers collapse several different phenomena into one comparison: retrieval versus no retrieval. In these experiments, that comparison is too coarse. LegalBench-SCALR loses heavily at top-1 and recovers at top-5, so a route that expands the candidate set is plausible. BarExamQA is top-k flat but improves with HyRE, so the route should preserve an answer frame rather than simply retrieve more passages. HousingQA likely requires metadata-constrained retrieval checks, because the question state is part of the legal problem and a normalized state filter now beats generic top-5 retrieval. CaseHOLD improves gold retrieval under two-call HyRE but not answer accuracy, so the route should focus on mapping retrieved holdings back to answer options.

This gives a feasible version of the "agentic" idea without making the agent itself the unit of evaluation. The method we are working toward is an adaptive HyRE controller: keep the snap-first reasoning step, but decide how it should sit on top of the problem's bottleneck. Instead of asking a planner to invent an open-ended research workflow, we can ask a controller to choose among a small set of evidence-budgeted interventions. The controller would observe cheap features before or during retrieval: question type, answer format, available metadata, top-score margin, retrieved-document agreement, candidate-option similarity, and whether the snap answer agrees with retrieval. It would then choose how to use the HyRE reasoning budget: ordinary RAG when retrieval is already adequate, top-k expansion when candidate recall is binding, HyRE when query framing is binding, diverse HyDE passages when one generated representation is too narrow, state-filtered retrieval when metadata is legally decisive, or option-aware reranking when retrieved evidence is not mapping back to choices. The result is still agentic in the sense that method adaptation happens on the fly, but it is experimentally cleaner because every route corresponds to a measured ablation cell.

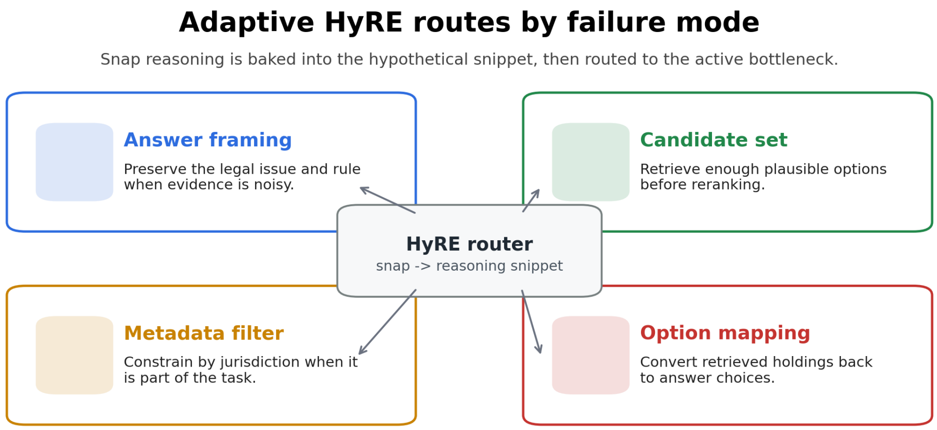

Figure 6: Conceptual adaptive HyRE routes. The route changes with the observed failure mode: answer framing, candidate-set expansion, metadata filtering, or option mapping. Figure 7 grounds these routes in the current benchmark evidence.

This framing also explains why the current negative results are useful. A negative CaseHOLD answer delta after a positive gold-hit delta is not just a negative method comparison; it identifies a downstream conversion problem. A flat BarExamQA top-k sweep does not make retrieval irrelevant; it says that retrieval depth is not the active lever for that task. HousingQA shows that jurisdiction metadata can matter as much as semantic similarity, while SCALR and CaseHOLD show that holding-selection tasks can be candidate-limited or conversion-limited in different ways. These are the hooks for a stronger final project: tune HyRE quality where query formulation is binding, tune query number or diversity where one hypothetical passage is too narrow, tune metadata filters where jurisdiction is binding, and tune verifier or option-mapping logic where evidence conversion is binding. The stubborn cases are therefore not outside the framing; they define which HyDE-adjacent component still needs to be designed. The current evidence does not yet instantiate the full adaptive controller, but it shows why that controller is the right next object to build: different legal tasks need different forms of retrieval-time reasoning, and the harness can measure which form is helping.

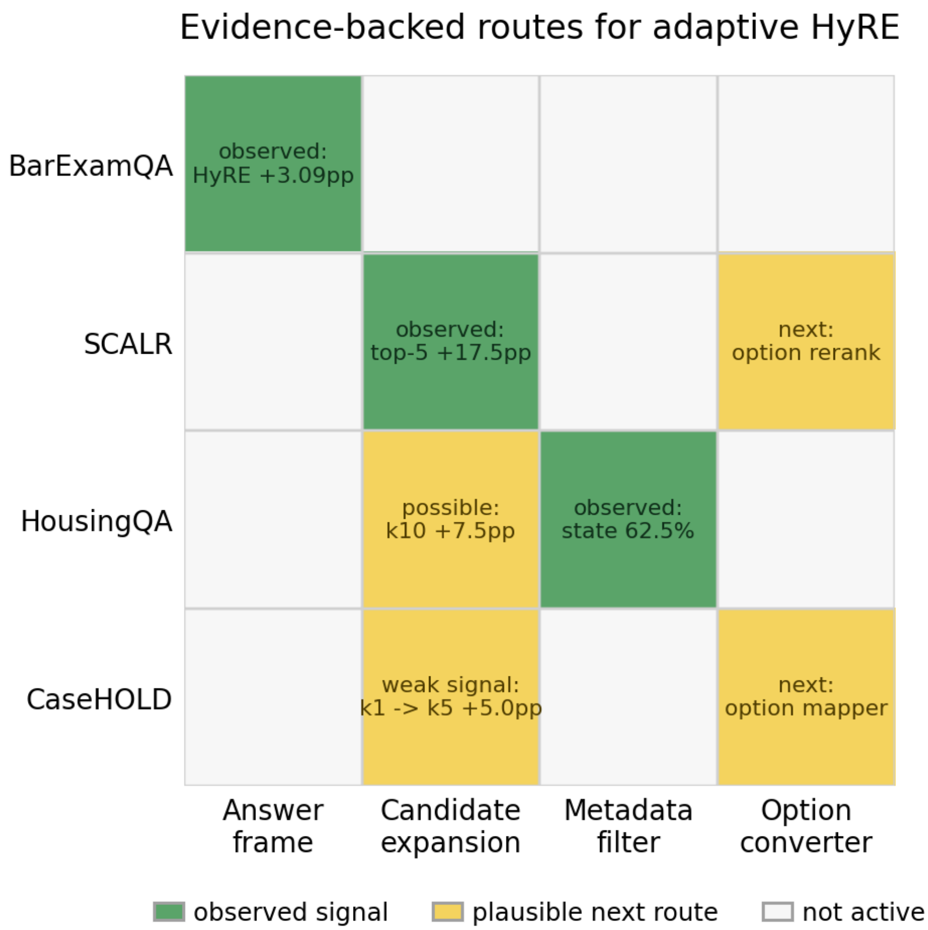

Figure 7: Route map implied by current evidence. Green cells are supported by observed route evidence; amber cells are plausible next interventions. The diagnostic protocol generalizes, while the current effects remain scoped to the model, task, and sample sizes reported above.

The resulting claim is deliberately scoped but optimistic. Across several legal benchmarks, we found one full-corpus legal HyRE lift on BarExamQA and several $N = 200$ diagnostic signals that expose reusable bottlenecks; those signals define a routing agenda. That is a better foundation for future work than another broad sweep over loosely comparable prompts, because it says what should change when a method fails.

7 Limitations and Future Work

The main limitation is statistical scope. Only BarExamQA has a full-corpus legal result; HousingQA, SCALR, and CaseHOLD are paired $N = 200$ diagnostic subsets for mechanism discovery rather than final benchmark rankings. These subsets are sufficient to motivate route choices, but larger evaluations are needed to test whether the same bottleneck patterns persist.

Future work should compare against full Self-RAG, CRAG, and Speculative RAG implementations once the legal routing policy is more stable. The next experiments should stay legal-only: test SCALR option grounding or reranking, test CaseHOLD option-aware synthesis, and add LegalBench-RAG or Legal RAG Bench for broader legal coverage. The adaptive HyRE controller should be evaluated by scaling the strongest routes first: run larger HousingQA state-filter evaluations, add SCALR option reranking after top-5 retrieval, and test CaseHOLD option-aware conversion.

8 Conclusion

The legal-only evidence supports a bottleneck-typed view of RAG. HyRE helps BarExamQA at full corpus, but BarExamQA is not retrieval-depth limited. SCALR is candidate-depth limited, HousingQA shows a real metadata-filtering signal, and CaseHOLD shows that better generated-query retrieval does not automatically convert to better answers. This is the project signal: adaptive legal RAG should not ask whether HyDE is globally good or bad; it should ask where the task's bottleneck is and apply the HyRE reasoning budget there. That gives us a concrete and testable path from the current report to a more general method: diagnose the bottleneck, select the route, and scale the paired evaluation once the route has signal.

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